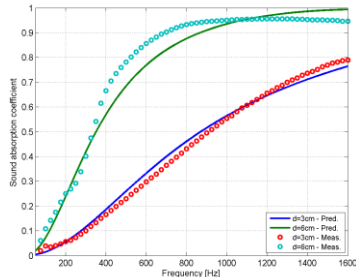
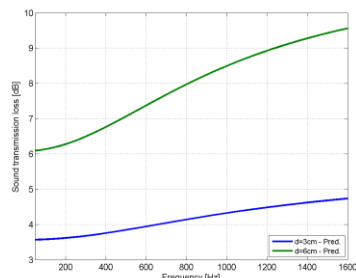


Acoustical database of porous materials

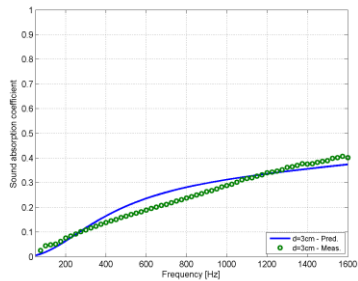
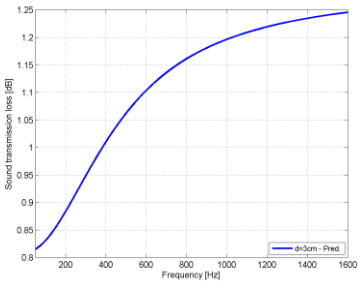
This document summarises in a database format the acoustical measurements of several porous materials.

Fibrous/Foam materials

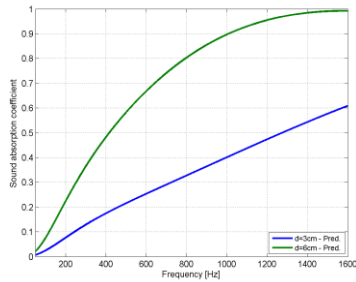
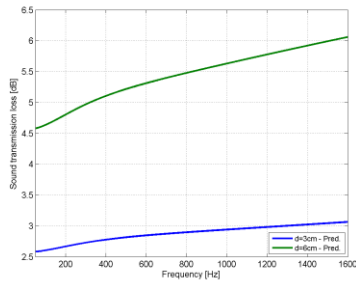
Isover - Industriplade 1

Material type	Glass wool
Brand	Isover
Material model	Industriplade 1
Code name	'GW' or 'GlassWool'
Meas. file name	GW_Analyzed.mat
Method used	Fitting routine - Differential evolution - low-freq
Pred. model	UTV perpendicular flow with JCAL
ϕ	0.9882
a [μm]	6.472
$k_0 \times 10^{10}$ [m^2]	13.1481
Λ [μm]	274.1991
α_∞	1.0118
$k'_0 \times 10^{10}$ [m^2]	26.2963
Λ' [μm]	542.0026
ρ_b [kg/m^3]	30
Picture	
SA - d=[3,6] cm	
TL - d=[3,6] cm	

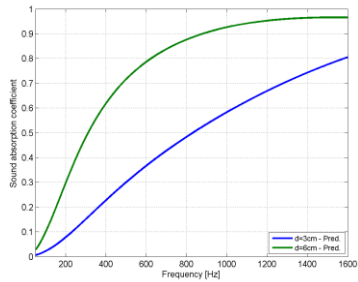
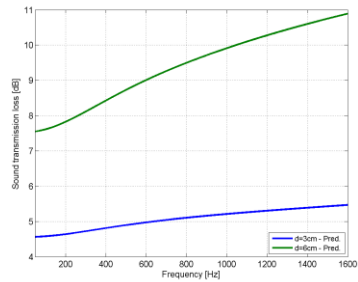
Merino wool

Material type	Merino wool
Brand	---
Material model	---
Code name	'MerinoWool'
Meas. file name	MerinoWool_Analyzed.mat
Method used	Fitting routine - Differential evolution - low-freq
Pred. model	UTV parallel flow with JCAL
ϕ	0.9841
a [μm]	12.742
$k_0 \times 10^{10}$ [m^2]	68.2393
Λ [μm]	788.6416
α_∞	1
$k'_0 \times 10^{10}$ [m^2]	68.2393
Λ' [μm]	542.0026
ρ_b [kg/m^3]	20.7241
Picture	
SA - d=3 cm	
TL - d=3 cm	

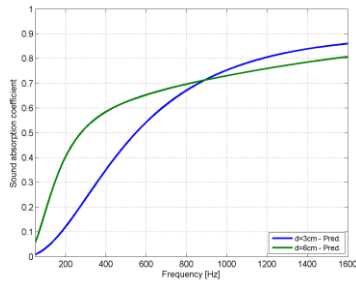
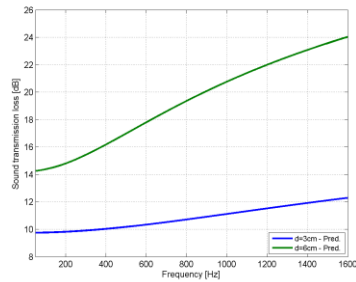
Isover_500

Material type	Mineral wool
Brand	Isover
Material model	Isover_500
Code name	'SGA' or ' Isover_500'
Meas. file name	Meas. taken by SG (data not available)
Method used	---
Pred. model	JCA
ϕ	0.977
a [μm]	---
$k_0 \times 10^{10}$ [m^2]	19.3360
Λ [μm]	145
α_∞	1
$k'_0 \times 10^{10}$ [m^2]	82.5565
Λ' [μm]	260
ρ_b [kg/m^3]	19
Picture	---
SA - d=3 cm	
TL - d=3 cm	

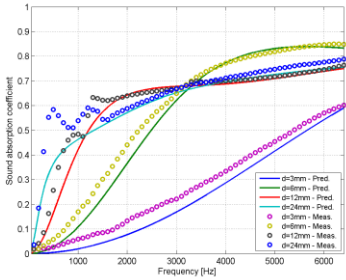
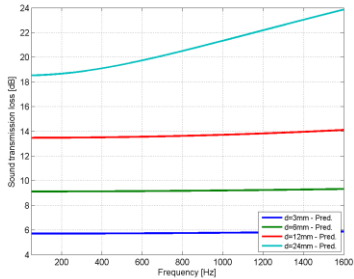
Isover_1000

Material type	Mineral wool
Brand	Isover
Material model	Isover_1000
Code name	'SGB' or ' Isover_1000'
Meas. file name	Meas. taken by SG (data not available)
Method used	---
Pred. model	JCA
ϕ	0.987
a [μm]	---
$k_0 \times 10^{10}$ [m^2]	9.6680
Λ [μm]	96
α_∞	1
$k'_0 \times 10^{10}$ [m^2]	44.5384
Λ' [μm]	190
ρ_b [kg/m^3]	33
Picture	---
SA - d=3,6 cm	
TL - d=3,6 cm	

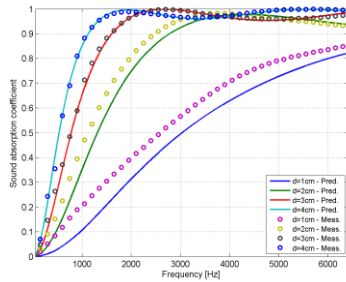
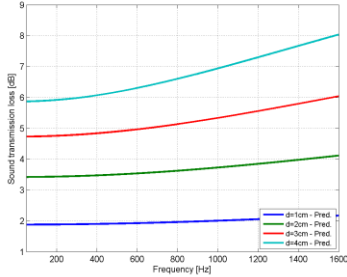
Ecophon

Material type	Mineral wool
Brand	Ecophon
Material model	---
Code name	'SGC' or ' Ecophon'
Meas. file name	Meas. taken by SG (data not available)
Method used	---
Pred. model	JCA
ϕ	0.97
a [μm]	---
$k_0 \times 10^{10}$ [m^2]	3.2227
Λ [μm]	27
α_∞	1.02
$k'_0 \times 10^{10}$ [m^2]	13.6237
Λ' [μm]	106
ρ_b [kg/m^3]	77
Picture	---
SA - d=3,6 cm	
TL - d=3,6 cm	

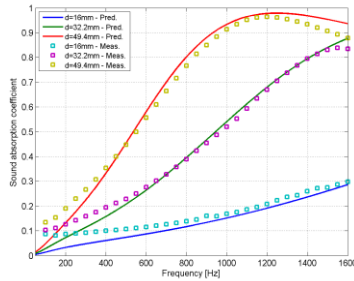
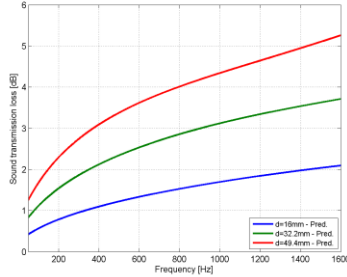
Dyson felt

Material type	Felt
Brand	Dyson felt
Material model	---
Code name	'Dyson_felt'
Meas. file name	Dyson_felt_Analyzed.MAT
Method used	Fitting routine - Differential evolution - low-freq (W), high-freq (S), combined low+high (WS)
Pred. model	JCA
ϕ	0.92
a [μm]	---
$k_0 \times 10^{10}$ [m^2]	0.7180
Λ [μm]	5.6471
α_∞	1.0014
$k'_0 \times 10^{10}$ [m^2]	2.6967
Λ' [μm]	48.4247
ρ_b [kg/m^3]	---
Picture	
SA - d=3,6,12,24 mm	
TL - d=3,6,12,24 mm	

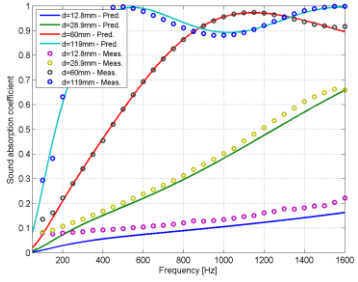
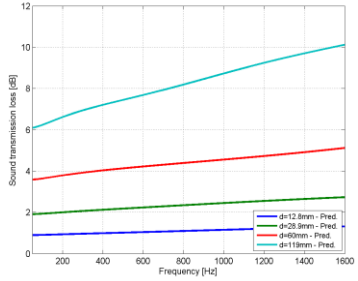
JLR felt

Material type	Felt (or fleece they call it)
Brand	JLR felt
Material model	---
Code name	'JLR' or 'JLR_felt'
Meas. file name	JLR_felt_Analyzed.mat
Method used	Fitting routine - Differential evolution - high-freq (S)
Pred. model	JCAL
ϕ	0.9946
a [μm]	---
$k_0 \times 10^{10}$ [m^2]	9.2164
Λ [μm]	63.4983
α_∞	1.0045
$k'_0 \times 10^{10}$ [m^2]	9.2180
Λ' [μm]	3151.7
ρ_b [kg/m^3]	---
Picture	
SA - d=1,2,3,4 cm	
TL - d=1,2,3, 4 cm	

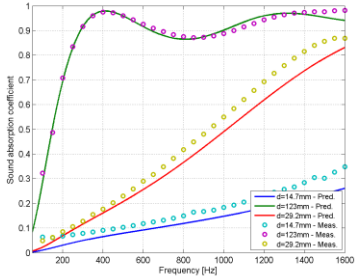
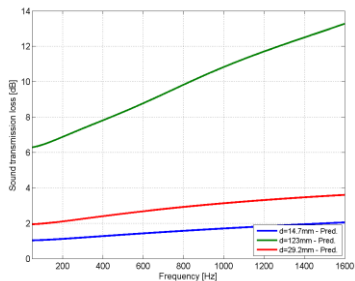
Foam filter (wrapping foam)

Material type	Wrapping foam (black)
Brand	Vicoustic - Unknown
Material model	---
Code name	'FoFi'
Meas. file name	FoFi_Analyzed.mat
Method used	Fitting routine - Differential evolution - low-freq (W)
Observation	Structure's got fields G1 (sample group 1), G2 (sample group 2) and G1_G2 (samples of group 2 mounted on samples of group 1)
Pred. model	JCAL
ϕ	0.9915
a [μm]	---
$k_0 \times 10^{10}$ [m^2]	111.161
Λ [μm]	63.4495
α_∞	1.152
$k'_0 \times 10^{10}$ [m^2]	123.425
Λ' [μm]	222.834
ρ_b [kg/m^3]	24.7056
Picture	
SA - d=16,32,48 mm (nominal)	
TL - d=16,32,48 mm (nominal)	

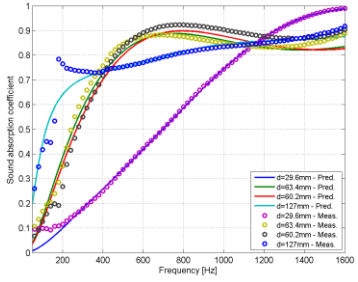
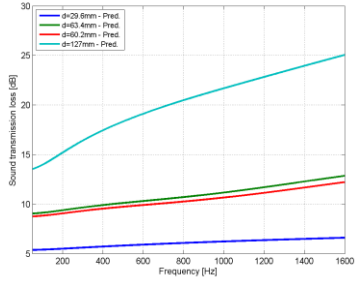
Vicycle low-density foam - ViLDFo

Material type	Recycled foam
Brand	Vicoustic
Material model	Vicycle - low-density foam
Code name	'ViLDFo'
Meas. file name	ViLDFo_Analyzed.mat
Method used	Fitting routine - Differential evolution - low-freq (W)
Pred. model	JCAL
ϕ	0.9894
a [μm]	---
$k_0 \times 10^{10}$ [m^2]	26.336
Λ [μm]	126.5925
α_∞	1.3659
$k'_0 \times 10^{10}$ [m^2]	102.559
Λ' [μm]	265.5769
ρ_b [kg/m^3]	51.9564
Picture	
SA - d (see legend)	
TL - d (see legend)	

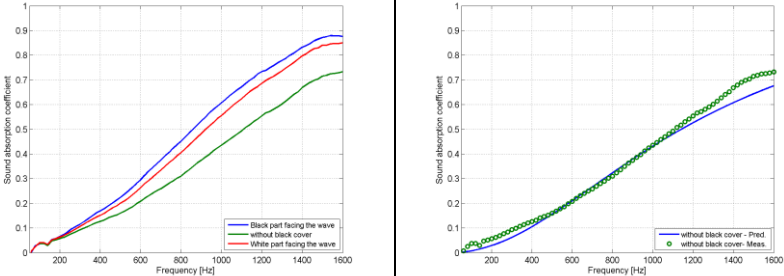
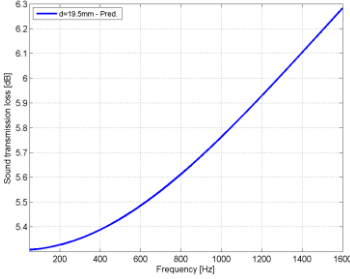
Vicycle medium-density foam - ViMDFo

Material type	Recycled foam
Brand	Vicoustic
Material model	Vicycle - medium-density foam
Code name	'ViMDFo'
Meas. file name	ViMDFo_Analyzed.mat
Method used	Fitting routine - Differential evolution - low-freq (W)
Pred. model	JCAL
ϕ	0.9811
a [μm]	---
$k_0 \times 10^{10}$ [m^2]	26.145
Λ [μm]	82.3756
α_∞	1.412
$k'_0 \times 10^{10}$ [m^2]	83.379
Λ' [μm]	211.787
ρ_b [kg/m^3]	92.6078
Picture	
SA - d (see legend)	
TL - d (see legend)	

Vicycle high-density foam - ViHDFo

Material type	Recycled foam
Brand	Vicoustic
Material model	Vicycle - high-density foam
Code name	'ViHDFo'
Meas. file name	ViHDFo_Analyzed.mat
Method used	Fitting routine - Differential evolution - low-freq (W)
Pred. model	JCAL
ϕ	0.9749
a [μm]	---
$k_0 \times 10^{10}$ [m^2]	7.7340
Λ [μm]	47.0902
α_∞	1.6059
$k'_0 \times 10^{10}$ [m^2]	76.814
Λ' [μm]	273.7406
ρ_b [kg/m^3]	122.7899
Picture	
SA - d (see legend)	
TL - d (see legend)	

3M Thinsulate

Material type	Fibrous
Brand	3M
Material model	Thinsulate
Code name	'Thinsulate'
Meas. file name	Thinsulate_Analyzed.mat
Method used	Fitting routine - Differential evolution - low-freq (W) - applied to material without the black cover
Pred. model	UTV perpendicular flow with JCAL
ϕ	0.991
a [μm]	3.3919
$k_0 \times 10^{10}$ [m^2]	5.1589
Λ [μm]	188.4248
α_∞	1.009
$k'_0 \times 10^{10}$ [m^2]	10.3178
Λ' [μm]	373.4882
ρ_b [kg/m^3]	---
Picture	
SA - $d=19.54 \pm 0.35$ mm	
TL - $d=19.54 \pm 0.35$ mm	

Granular materials

SRD06

Material type	Granular
Brand	Chemviron
Material model	SRD06
Code name	'SRD06'
Meas. file name	SRD06_Analyzed.mat
Method used	Fitting routine - Differential evolution - See Venegas' PhD thesis
Pred. model	Venegas' model
ϕ_p	0.332
r_p [mm]	0.3062
ϕ_m	0.664
r_m [μm]	0.1887
ϕ_n	0.2791
r_n [nm]	2.4278
k_a [1/Pa/s]	0.0369
k_d [1/s]	48.0050
ρ_b [kg/m ³]	---
Picture	
SA - d=5, 30 mm	
TL - d=5, 30 mm	

Repeat this with all the activated carbon samples